



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

YEAR 11

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 1**

Mathematics Extension 1

General Instructions

- Working time – 90 minutes.
- Reading Time – 5 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work

Total Marks –80

- Attempt all questions.
- All questions are **NOT** of equal value.

Examiner: *C. Kourtesis*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Question 1 (15 marks)

Marks

(a) Simplify $\frac{1}{5!} + \frac{1}{6!}$ 1

(b) In how many ways can a committee of 6 be chosen from a group of 10 people? 2

(c) Write down the general solution of $\cos \theta = \frac{1}{2}$ 2

(d) If α, β and γ are the roots of the cubic equation $2x^3 + 12x^2 - 6x + 1 = 0$ find the value of: 4

(i) $\alpha + \beta + \gamma$

(ii) $\alpha\beta\gamma$

(iii) $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$

(e) The equation of a parabola is 3

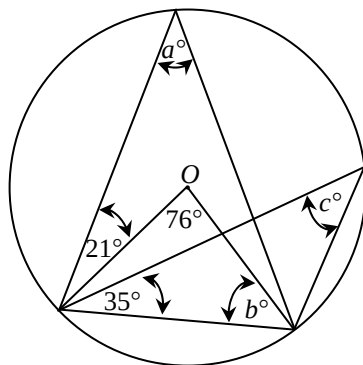
$$y = x^2 - 6x + 7$$

Find the

(i) focal length

(ii) coordinates of the vertex

(f) 3



Find the values of a , b and c .
[There is no need to give reasons]

O is the centre of the circle

Question 2 [14 marks]**Marks**

- (a) The polynomial $P(x) = x^3 - 6x^2 + \theta x - 4$ has $x = 1$ as a zero.

5

Find the:

- (i) value of θ
- (ii) other zeros of $P(x)$
- (iii) values of x for which $P(x) < 0$

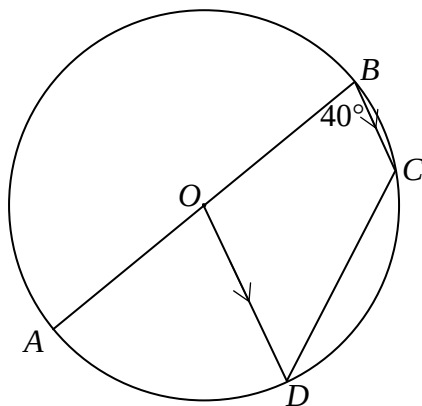
- (b) The parametric coordinates of P , a point on a curve are given by $P(8t, 2t^2)$ where t is the parameter

3

Find:

- (i) the Cartesian equation of the curve
- (ii) the gradient of the tangent to the curve at the point where $t = 3$

- (c)



AB is the diameter of a circle centre O .
 BC and OD are parallel and $\angle OBC = 40^\circ$. Find the size of $\angle OCD$ giving reasons

3

- (d) Prove by Mathematical Induction that

4

$$1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$$

for positive integers n

Question 3**[18 marks]****Marks**

- (a) T is the point $(1, -2)$ and the tangents from T to the parabola $x^2 = 12y$ touch the parabola at A and B . Write down the equation of the line AB 2

- (b) In how many ways can 5 different books be placed in a row so that two specified books: 4

- (i) occupy the end positions
- (ii) must always be together?

- (c) (i) Write down the expansion of $\sin(x - \theta)$ 6

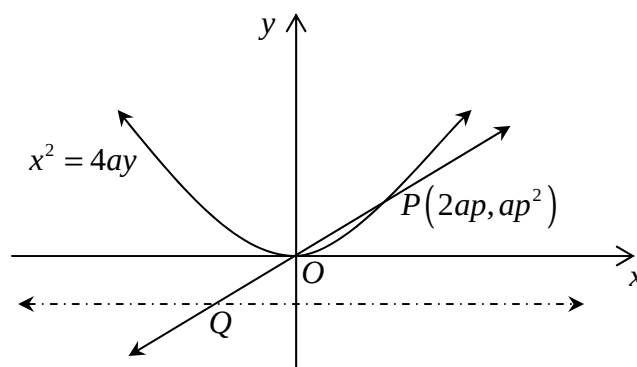
- (ii) If $\sin x - \cos x = A \sin(x - \theta)$ where $A > 0$ and $0 < \theta < \frac{\pi}{2}$, show that $A = \sqrt{2}$ and $\theta = \frac{\pi}{4}$

- (iii) Hence or otherwise solve the equation for **all** values of x .

$$\sin x - \cos x = 1$$

- (iv) Find the maximum value of $\sin x - \cos x$

- (d) 6



The diagram shows the parabola $x^2 = 4ay$. PO is produced to meet the directrix at Q .

- (i) Show that the equation of the tangent at P has equation $y - px + ap^2 = 0$
- (ii) Find the coordinates of the point Q .
- (iii) Prove that QS is parallel to the tangent at P .
[S is the focus]

Question 4	[18 marks]	Marks
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- | | | |
|-----|---|---|
| (a) | 17 people sit at a round table. In how many ways can they be seated if: | 4 |
| | (i) there are no restrictions | |
| | (ii) two particular people cannot sit together | |

- | | | |
|-----|--|---|
| (b) | If $t = \tan \frac{\theta}{2}$ express in terms of t | 3 |
|-----|--|---|

$$\cos \theta + \sin^2 \left(\frac{\theta}{2} \right)$$

- | | | |
|-----|---|---|
| (c) | When a polynomial $P(x)$ is divided by $x^2 - 3x + 2$ the remainder is $4x - 7$.
Find the remainder when $P(x)$ is divided by $(x - 1)$. | 2 |
|-----|---|---|

- | | | |
|-----|---|---|
| (d) | A polynomial $P(x)$ has the following properties: | 4 |
| | (i) $P(x)$ is odd and has a factor of $(x - 5)^2$. | |
| | (ii) The curve of $y = P(x)$ passes through $(1, 1152)$. | |

Find the polynomial, $P(x)$ of least degree that satisfies the above, expressing your answer in factorized form.

- | | | |
|-----|--|---|
| (e) | $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$ with parameters p and q respectively. | 5 |
| | (i) Find the coordinates of M the midpoint of PQ . | |
| | (ii) If PQ subtends a right angle at the vertex show that $pq + 4 = 0$. | |
| | (iii) Find the locus of M . | |

Question 5**[15 marks]****Marks**

- (a) How many arrangements are there of the letters in the word

2

ANDAMOOKA

- (b) From the letters of the word PROBLEMS how many different words consisting of 5 letters are possible if they include P, do not begin with P and the letter M is to be excluded?

3

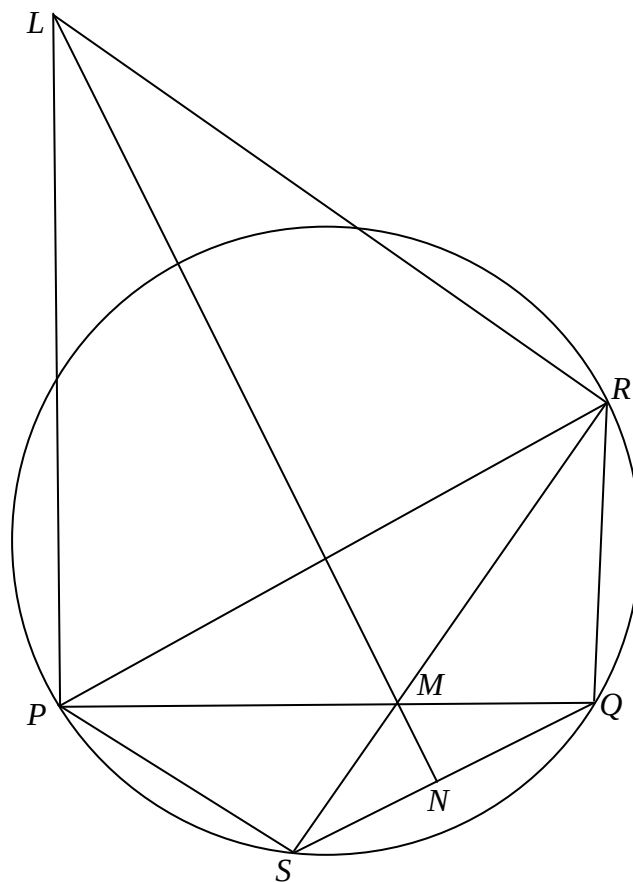
- (c) Prove by Mathematical Induction that

4

$$\sin(n\pi + \theta) = (-1)^n \sin \theta \text{ for } 0 < \theta < \frac{\pi}{2} \text{ for positive integers } n$$

- (d) PQ and RS are two chords of a circle which intersect at M inside the circle. MN is the perpendicular from M to SQ . L is the point on NM produced such that LP is perpendicular to PQ

6



- (i) Copy the diagram
- (ii) Show that $\triangle PML \parallel \triangle NMQ$
- (iii) Hence show that $LR \perp RS$

End of paper



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Mathematics Extension 1

Sample Solutions

Question	Marker
1	Mr Kidd
2	Mr Choy
3	Mr Parker
4	Mr Dunne
5	Mr Bigelow

Question 1

$$\frac{a}{3} (a) \frac{1}{5!} + \frac{1}{6!} = \frac{6}{6!} + \frac{1}{6!}$$

$$= \frac{7}{6!} \text{ or } \frac{7}{720}$$

$$(b) {}^{10}C_6 = 210$$

$$(c) \theta = 2n\pi \pm \frac{\pi}{3} \text{ or } 360^\circ n \pm 60^\circ$$

$$(d)(i) \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{12}{2} = -6$$

$$(ii) \alpha \beta \gamma = -\frac{d}{a}$$

$$= -\frac{1}{2}$$

$$(iii) \alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{-\frac{6}{2}}{-\frac{1}{2}}$$

$$= 6$$

$$(e) y = x^2 - 6x + 7$$

$$\text{ie } x^2 - 6x + 9 = y - 7 + 9$$

$$(x-3)^2 = y+2$$

$$(x-3)^2 = 4 \times \frac{1}{4} (y+2)$$

$$\therefore \text{Focal length} = \frac{1}{4}$$

$$\text{Vertex is } (3, -2)$$

$$(f) a = \frac{76}{2} = 38^\circ$$

$$b = \frac{180-76}{2}$$

$$= 52^\circ$$

$$c = a = 38^\circ$$

(a)

$$p(x) = x^3 - 6x^2 + 9x - 4$$

$$p(1) = 0$$

$$1 - 6 + 9 - 4 = 0$$

$$9 - 9 = 0$$

$$(i) \therefore 9 = 9$$

$$(ii) \frac{x^2 - 5x + 4}{x-1}$$

$$x-1 \overline{) x^3 - 6x^2 + 9x - 4}$$

$$-(x^3 - x^2)$$

$$-5x^2 + 9x - 4$$

$$-(-5x^2 + 5x)$$

$$4x - 4$$

$$-(4x - 4)$$

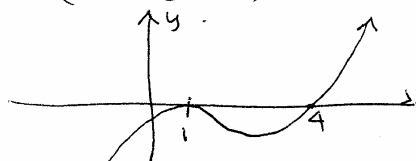
$$0$$

$$\therefore p(x) = (x-1)^2(x-4)$$

$$x = 1, 4$$

$$(iii) p(x) < 0$$

$$(x-1)^2(x-4) < 0$$



from graph

$$x < 4 \quad x \neq 1$$

$$(b) \quad x = 8t, \quad y = 2t^2$$

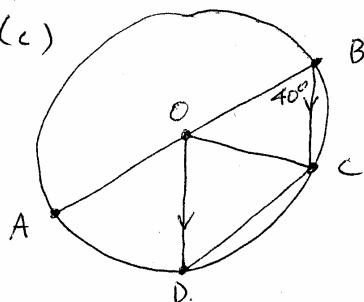
$$y = 2\left(\frac{x}{8}\right)^2 = \frac{x^2}{32}$$

$$(i) \therefore x^2 = 32y$$

$$(ii) \quad \frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt} = 4t/8 = t/2$$

$$\therefore \frac{dy}{dx} \Big|_{t=3} = \frac{3}{2}$$

(c)



Join OC.

$\triangle OBC$ isosceles
($OB = OC$, radii)

$$\therefore \angle BCO = 40^\circ$$

Also, $\angle COD = 40^\circ$

(alt. $\angle$ s, $BC \parallel OD$).

but $\triangle ODC$ is also isosceles.

$$OC = OD \text{ (radii)}$$

$$\Rightarrow \angle OCD = 70^\circ$$

($\angle$ sum of a triangle ODC).

(d) Let $S(n)$ be the proposition that

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

For $n=1$, LHS = 1

$$RHS = 1^2 = 1$$

$\therefore LHS = RHS \Rightarrow S(1)$ is true.

Assume $S(k)$ is true

i.e.

$$1 + 3 + 5 + \dots + (2k-1) = k^2$$

Consider $n = k+1$

$$1 + 3 + \dots + (2k-1) + (2k+1)$$

$$= k^2 + (2k+1)$$

$$= (k+1)^2$$

Since the statement is true for $n=1$ and true for $n=k+1$ when true for $n=k$ ($k \in \mathbb{Z}^+$) $\therefore S(n)$

is true for all $n \geq 1$.

Question 3

(a) Chord of contact $xx_0 = 2a(y + y_0)$

$$x^2 = 12y \Rightarrow a = 3$$

$$T\left(\underset{x_0}{1}, \underset{y_0}{-2}\right)$$

$$\therefore x \times 1 = 2 \times 3(y - 2) \Rightarrow x = 6(y - 2)$$

$$\therefore x - 6y + 12 = 0$$

(b) (i) 

Place the two books at either end, so there are 3! ways of arranging the books in between. The end books can then be switched.

$$\text{Total} = 3! \times 2 = 12$$

(ii) 

Place the two books together, so there are 4 objects to arrange in 4! Ways. Then the books can be switched.

$$\text{Total} = 4! \times 2 = 48$$

(c) (i) $\sin(x - \theta) = \sin x \cos \theta - \cos x \sin \theta$

$$\begin{aligned} \text{(ii)} \quad \sin x - \cos x &= A \sin(x - \theta) \\ &= (A \cos \theta) \sin x - (A \sin \theta) \cos x \end{aligned}$$

$$\therefore A \cos \theta = 1, A \sin \theta = 1$$

$$\therefore A = \sqrt{1^2 + 1^2} = \sqrt{2} \quad (\because A > 0)$$

$$\tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4} \quad \left(\because 0 < \theta < \frac{\pi}{2} \right)$$

$$\text{(iii)} \quad \sin x - \cos x = 1 \Rightarrow \sqrt{2} \sin\left(x - \frac{\pi}{4}\right) = 1$$

$$\therefore \sin\left(x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$\boxed{\sin \theta = c \Rightarrow \theta = n\pi + (-1)^n \sin^{-1} c}$$

$$\therefore x - \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{4} = n\pi + \frac{\pi}{4} (1 + (-1)^n)$$

$$x = \begin{cases} n\pi & n \text{ odd} \\ n\pi + \frac{\pi}{2} & n \text{ even} \end{cases}$$

Alternatively:

$$\sin\left(x - \frac{\pi}{4}\right) = \cos\left(\frac{3\pi}{4} - x\right) = \cos\left(x - \frac{3\pi}{4}\right)$$

$$\therefore x - \frac{3\pi}{4} = 2n\pi \pm \frac{\pi}{4}$$

$$\cos \theta = c \Rightarrow \theta = 2n\pi \pm \cos^{-1} c$$

$$x = 2n\pi + \pi, 2n\pi + \frac{\pi}{2}$$

$$\therefore x = (2n+1)\pi, 2n\pi + \frac{\pi}{2}$$

$$(iv) \quad \max(\sin x - \cos x) = \max\left(\sqrt{2} \sin\left(x - \frac{\pi}{4}\right)\right) = \sqrt{2}$$

$$(d) \quad (i) \quad y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

$$m_p = \frac{dy}{dx}\bigg|_{x=2ap} = \frac{2ap}{2a} = p$$

$$\therefore y - ap^2 = p(x - 2ap)$$

$$\therefore y = px - 2ap^2 + ap^2 = px - ap^2$$

Alternatively:

$$x = 2at, y = at^2$$

$$\frac{dx}{dt} = 2a, \frac{dy}{dt} = 2at$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2at}{2a} = t$$

$$\therefore m_p = \frac{dy}{dx}\bigg|_{t=p} = p$$

(ii) The directrix is $y = -a$

$$m_{OP} = \frac{ap^2 - 0}{2ap - 0} = \frac{p}{2}$$

$$\therefore y - ap^2 = \frac{p}{2}(x - 2ap) = \left(\frac{p}{2}\right)x - ap^2$$

$$\therefore y = \left(\frac{p}{2}\right)x$$

$$Q: \text{sub } y = -a \Rightarrow x = -\frac{2a}{p}$$

$$Q\left(-\frac{2a}{p}, -a\right)$$

(iii) Focus is $(0, a)$

$$m_{QS} = \frac{a - (-a)}{0 - \left(-\frac{2a}{p}\right)} = \frac{2a}{2a/p} = 2a \times \frac{p}{2a} = p$$

$\therefore QS \parallel$ tangent at P .

Question 4

a) i) $16!$ Standard Bookwork.

ii) If 2 people MUST be seated together

Consider them as joined $A \& B$ or $B \& A$

This can be done in $2 \times 15!$ ways

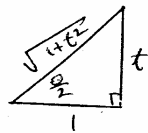
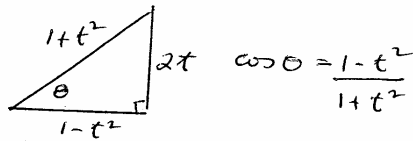
Hence 2 people not together $16! - 2 \times 15!$

$$= 15! (16-2)$$

$$= 14 \times 15!$$

=

b)



$$\tan \frac{\theta}{2} = t$$

$$\text{Hence } \sin \frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}}$$

$$\sin^2 \frac{\theta}{2} = \frac{t^2}{1+t^2}$$

$$\cos \theta + \sin^2 \frac{\theta}{2} = \frac{1-t^2}{1+t^2} + \frac{t^2}{1+t^2} = \frac{1}{1+t^2}$$

$$\begin{aligned} \text{c) } P(x) &= (x^2 - 3x + 2) Q(x) + (4x - 7) \\ &= (x-1)(x-2) Q(x) + (4x-7) \\ P(1) &= 0 + 4 - 7 \\ &= -3 \end{aligned}$$

d) If $P(x)$ is odd then curve passes through origin
ie x is a factor

If $(x-5)^2$ is a factor then $(x+5)^2$ must also be
a factor $P(-5) = -P(5) = 0$

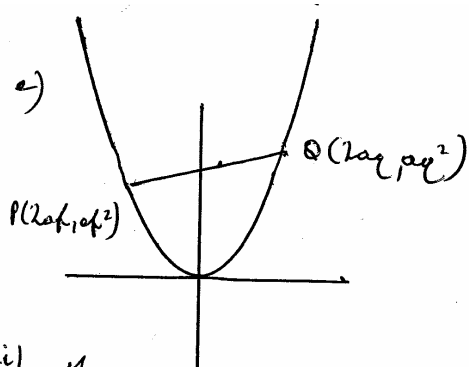
$$\text{Hence } P(x) = kx(x-5)^2(x+5)^2$$

$$\text{If } P(1) = 1152$$

$$1152 = k \times 16 \times 36$$

$$k = 2$$

$$\text{Hence } P(x) = 2x(x-5)^2(x+5)^2$$



i) M $x = \frac{2af + 2aq}{2}$
 $= a(p+q) \quad \text{--- (1)}$

$$y = \frac{af^2 + aq^2}{2}$$

$$= \frac{a}{2} (p^2 + q^2) \quad \text{--- (2)}$$

ii) Gradient OP = $\frac{af^2}{2af} = \frac{f}{2}$

Gradient OQ = $\frac{q}{2}$

Since $\perp$ $\frac{f}{2} \cdot \frac{q}{2} = -1$
 $fq + 4 = 0$

iii) From (1) $\frac{x^2}{a^2} = p^2 + 2pq + q^2$
 $= p^2 + q^2 - 8$

$$p^2 + q^2 = \frac{x^2}{a^2} + 8$$

From (2)

$$p^2 + q^2 = \frac{2y}{a}$$

Hence $\frac{2y}{a} = \frac{x^2}{a^2} + 8$

$$y = \frac{x^2}{2a} + 4a$$

Question 5

QUESTIONS.

$$(a). \frac{9!}{3! \times 2!} = \boxed{30240}$$

(b) Place first letter in 6 ways. then place P, ie 4 ways.
then place the other three letters ie. $5 \times 4 \times 3$ ways.

$$\therefore 6 \times 4 \times 5 \times 4 \times 3 = \boxed{1440}$$

$$(c) \text{ when } n=1. \quad LHS = \sin(\pi + \theta) = -\sin \theta$$

$$RHS = (-1)^1 \sin \theta = -\sin \theta.$$

$\therefore$ True when $n=1$.

$$\text{when } n=k. \quad \sin(k\pi + \theta) = (-1)^k \sin \theta.$$

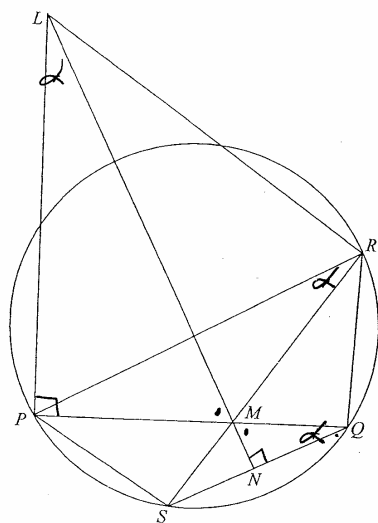
Assuming above to be true, show that
it is also true when $n=k+1$.

$$\text{ie } \sin((k+1)\pi + \theta) = (-1)^{k+1} \sin \theta.$$

$$\begin{aligned} LHS &= \sin((k+1)\pi + \theta) \\ &= \sin((k\pi + \theta) + \pi) \\ &= -\sin(k\pi + \theta) \quad (\text{using the identity } \sin(A+\pi) = -\sin A) \\ &= -1 \times (-1)^k \sin \theta \\ &= (-1)^{k+1} \sin \theta \\ &= RHS. \end{aligned}$$

Having assumed true for $n=k$ we proved true for $n=k+1$. Now since it is true for $n=1$ it is true for $n=2$ etc and hence true for all positive integers.

(d). (1)



(ii) $\angle LPM = \angle LQM = 90^\circ$ (data)
 $\angle LMP = \angle LMQ$ (vertically opposite)
 $\therefore \triangle PML \parallel \triangle QML$ (equiangular)

(iii) $\angle PLM = \angle QLM = \alpha$ (corresponding angles of similar triangles)
 $\angle LQM = \angle LSP = \alpha$ (angles in the same segment standing on the same arc are equal).
 $\therefore \angle PLM = \angle QLM = \alpha$.

$\therefore PMRL$ is a cyclic quadrilateral. (The interval PM subtends equal angles at the points L and R)

$\therefore \angle LRS = 90^\circ$ (opposite angles of a cyclic quadrilateral are supplementary)
 $\therefore LR \perp RS$.